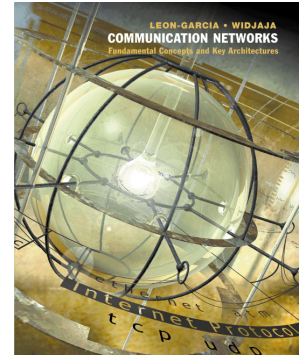


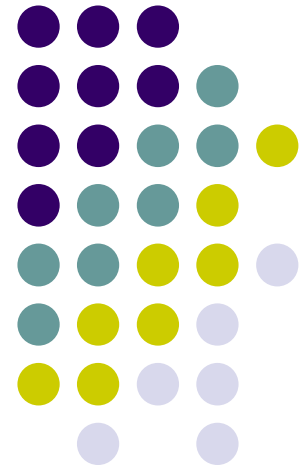
Lecture 5(Chapter 8)

Communication Networks and Services



Internet Routing Protocols

Part 2





Outline

- Datagrams *and Virtual Circuits*
- Basic Routing
- Routing Information Protocol (RIP)
- Open Shortest Path First (OSPF)
- Border Gateway Protocol (BGP)

Open Shortest Path First



- RFC 2328 (v2)
- Fixes some of the deficiencies in RIP
- Enables each router to learn complete network topology
- Each router monitors the *link state* to each neighbor and floods the link-state information to other routers
- Each router builds an identical *link-state database*
- Allows router to build shortest path tree with router as root
- OSPF typically converges faster than RIP when there is a failure in the network

OSPF Features



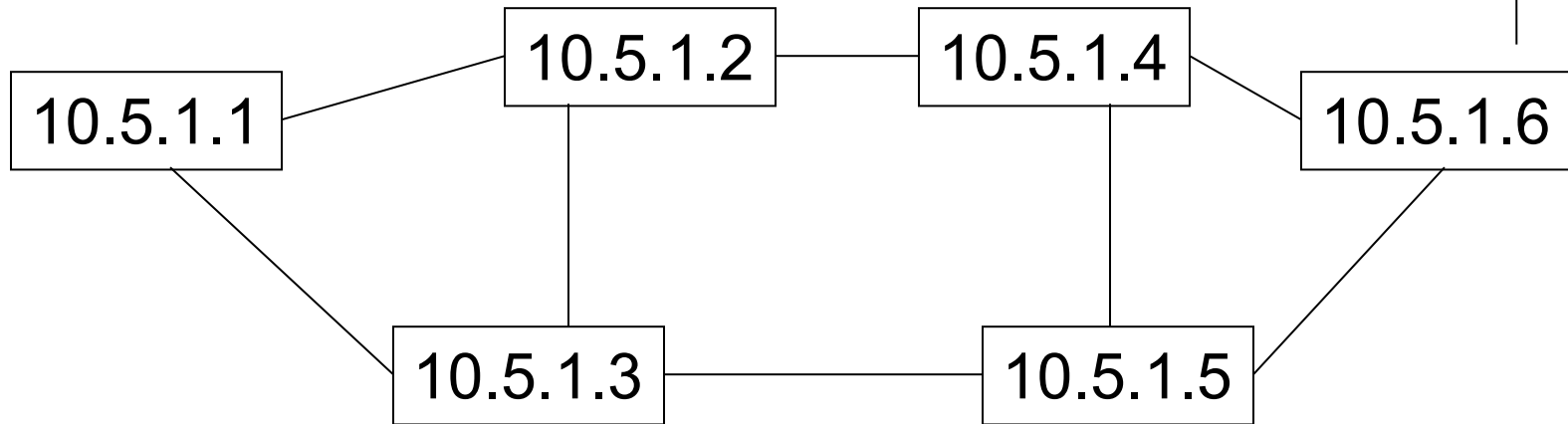
- *Multiple routes* to a given destination, one per type of service
- Support for *variable-length subnetting* by including the subnet mask in the routing message
- More *flexible link cost* which can range from 1 to 65,535
- Distribution of traffic over *multiple paths* of equal cost (load balance)
- *Authentication* to ensure routers exchange information with trusted neighbors
- *Designated router* to minimize table maintenance overhead



Flooding

- Used in OSPF to distribute link state (LS) information
- Forward incoming packet to all ports except where packet came in
- Packet eventually reaches destination as long as there is a path between the source and destination
- Generates exponential number of packet transmissions
- Approaches to limit # of transmissions:
 - Use a TTL at each packet; won't flood if TTL is reached
 - Each router adds its identifier to header of packet before it floods the packet; won't flood if its identifier is detected
 - Each packet from a given source is identified with a unique sequence number; won't flood if sequence number is same

Example OSPF Topology



At steady state:

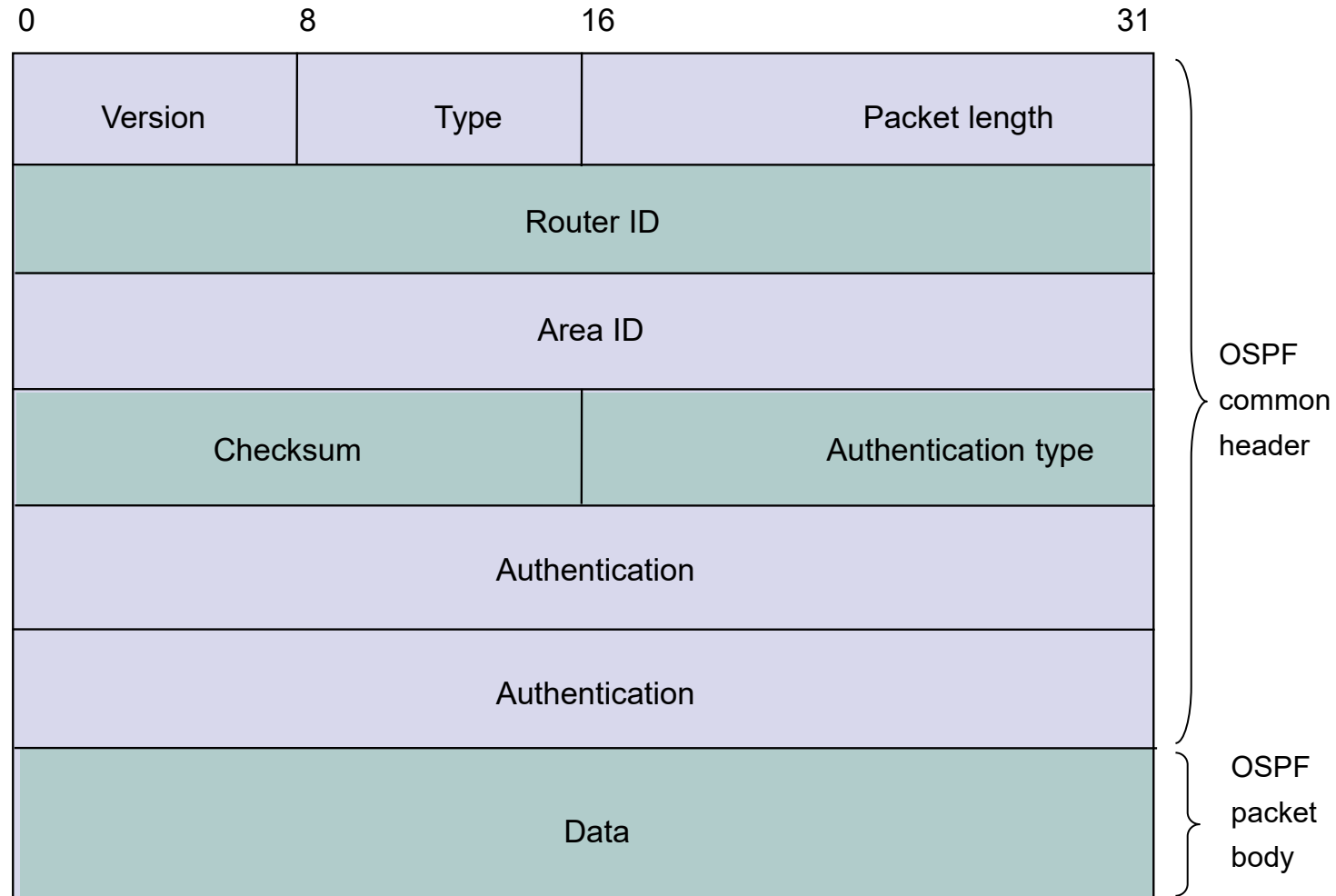
- All routers have same LS database
- Know how many routers in network
- Interfaces & links between routers
- Cost of each link
- Occasional Hello messages (10 sec) & LS updates sent (30 min)

OSPF Protocol



- OSPF packets transmitted directly on IP datagrams; Protocol ID 89
- OSPF packets sent to multicast address 224.0.0.5 (allSPFRouter on pt-2-pt)
- Five OSPF packet types:
 - *Hello*
 - *Database description*
 - *Link state request; Link state update; Link state ack*

OSPF Header



- Type: Hello, Database description, Link state request, Link state update, Link state acknowledgements



ospfcap - Ethereal

File Edit Capture Display Tools Help

No.	Time	Source	Destination	Protocol	Info
3	3.682599	20.1.1.1	224.0.0.5	OSPF	Hello Packet
6	6.189717	20.1.1.2	224.0.0.5	OSPF	Hello Packet
10	10.495667	20.1.1.1	224.0.0.5	OSPF	Hello Packet
11	11.264514	20.1.1.1	224.0.0.5	OSPF	LS Update
12	11.766168	20.1.1.2	224.0.0.5	OSPF	LS Update
13	11.775527	20.1.1.1	224.0.0.5	OSPF	LS Update
15	13.731086	20.1.1.2	224.0.0.5	OSPF	LS Acknowledge
16	13.740725	20.1.1.1	224.0.0.5	OSPF	LS Acknowledge
19	16.188992	20.1.1.2	224.0.0.5	OSPF	Hello Packet
21	19.018319	20.1.1.1	224.0.0.5	OSPF	LS Acknowledge
22	19.145138	20.1.1.2	224.0.0.5	OSPF	LS Acknowledge
24	20.493778	20.1.1.1	224.0.0.5	OSPF	Hello Packet
28	26.103357	20.1.1.2	224.0.0.5	OSPF	Hello Packet
33	30.407765	20.1.1.1	224.0.0.5	OSPF	Hello Packet

Frame 11 (226 bytes on wire, 226 bytes captured)

- Ethernet II, Src: 00:10:7b:39:ab:42, Dst: 01:00:5e:00:00:05
- Internet Protocol, Src Addr: 20.1.1.1 (20.1.1.1), Dst Addr: 224.0.0.5 (224.0.0.5)
- Open Shortest Path First
 - OSPF Header
 - LS Update Packet
 - Number of LSAs: 5
 - LS Type: Summary-LSA (IP network)
 - LS Age: 1 seconds
 - Options: 0x22 (E/DC)
 - Link-State Advertisement Type: Summary-LSA (IP network) (3)
 - Link State ID: 172.16.16.0
 - Advertising Router: 200.1.1.1 (200.1.1.1)
 - LS Sequence Number: 0x80000004
 - LS Checksum: abb5
 - Length: 28
 - Netmask: 255.255.255.0
 - Metric: 64
 - LS Type: Summary-LSA (IP network)
 - LS Type: AS-External-LSA (ASBR)
 - LS Type: AS-External-LSA (ASBR)
 - LS Type: AS-External-LSA (ASBR)

0000 01 00 5e 00 00 05 00 10 7b 39 ab 42 08 00 45 c0 ..^..... {9.B..E.
0010 00 d4 fa e9 00 00 01 59 c8 20 14 01 01 01 e0 00Y
0020 00 05 02 04 00 c0 c8 01 01 01 00 00 00 00 cc 96
0030 00 00 00 00 00 00 00 00 00 00 00 00 05 00 01
0040 22 03 ac 10 10 00 c8 01 01 01 80 00 00 04 ab b5 ".....

Filter: / Reset Apply File: ospfcap

Outline



- Basic Routing
- Routing Information Protocol (RIP)
- Open Shortest Path First (OSPF)
- Border Gateway Protocol (BGP)

Exterior Gateway Protocols

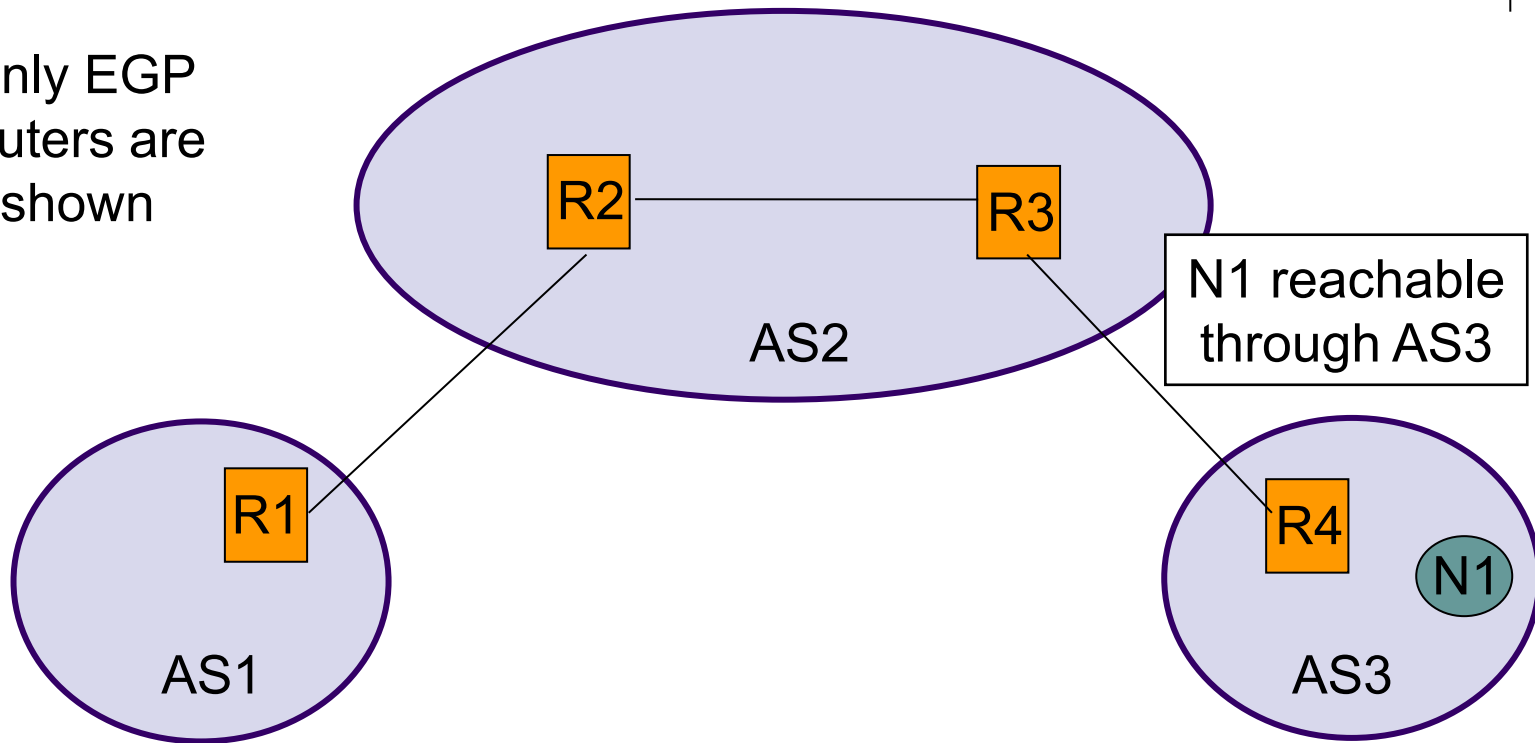


- Within each AS, there is a consistent set of routes connecting the constituent networks
- The Internet is woven into a coherent whole by *Exterior Gateway Protocols (EGPs)* that operate between AS's
- EGP enables two AS's to exchange routing information about:
 - The networks that are contained within each AS
 - The AS's that can be reached through each AS
- EGP path selection guided by policy rather than path optimality
 - Trust, peering arrangements, etc

EGP Example

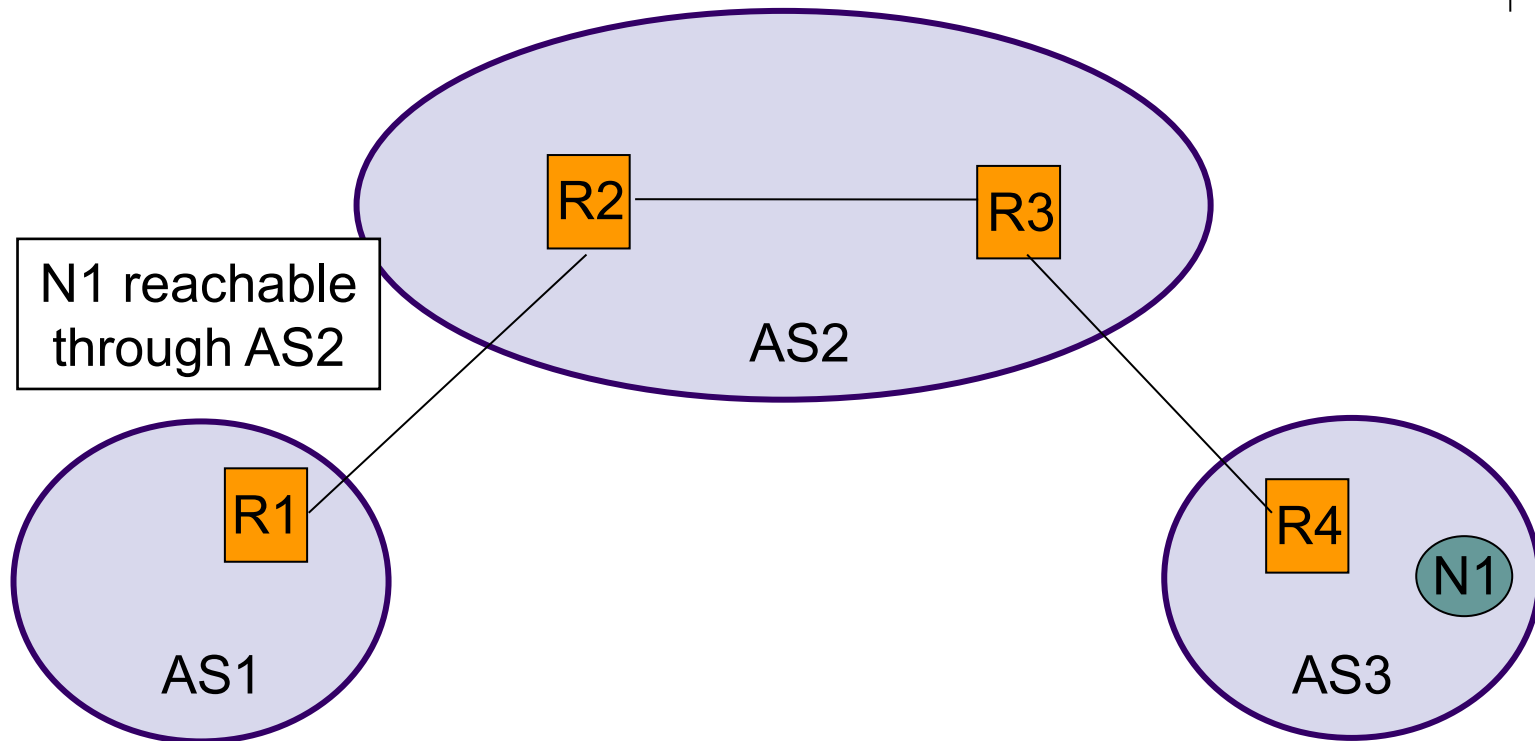


Only EGP
routers are
shown



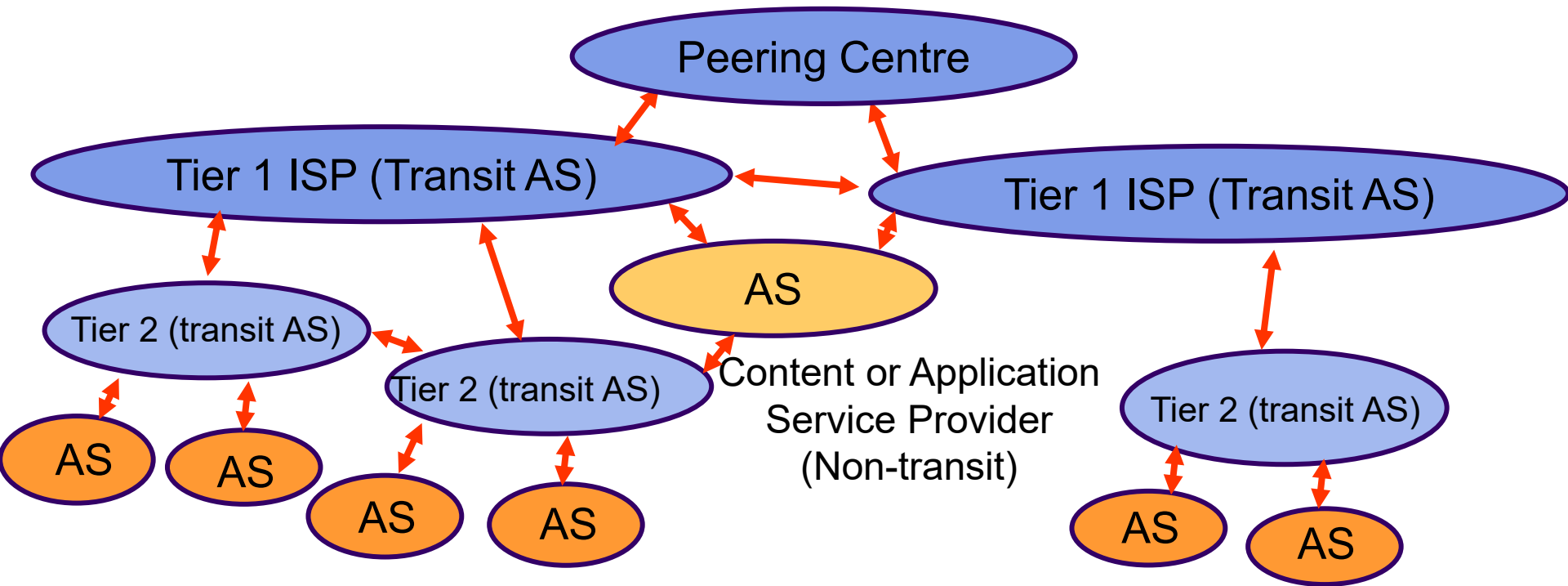
- R4 advertises that network N1 can be reached through AS3
- R3 examines announcement & applies *policy* to decide whether it will forward packets to N1 through R4
- If yes, routing table updated in R3 to indicate R4 as next hop to N1
- IGP propagates N1 reachability information through AS2

EGP Example



- EGP routers within an AS, e.g. R3 and R2, are kept consistent
- Suppose AS2 willing to handle *transit* packets from AS1 to N1
- R2 advertises to AS1 the reachability of N1 through AS2
- R1 applies its policy to decide whether to send to N1 via AS2

Peering and Inter-AS connectivity



- Non-transit AS's (stub & multihomed) do not carry transit traffic
- Tier 1 ISPs peer with each other, privately & peering centers
- Tier 2 ISPs peer with each other & obtain transit services from Tier 1s; Tier 1's carry transit traffic between their Tier 2 customers
- Client AS's obtain service from Tier 2 ISPs

Peering: OutDegree

AS Latency vs. OutDegree (n.p. 200 Quesant)

0

125

171

343

515

687

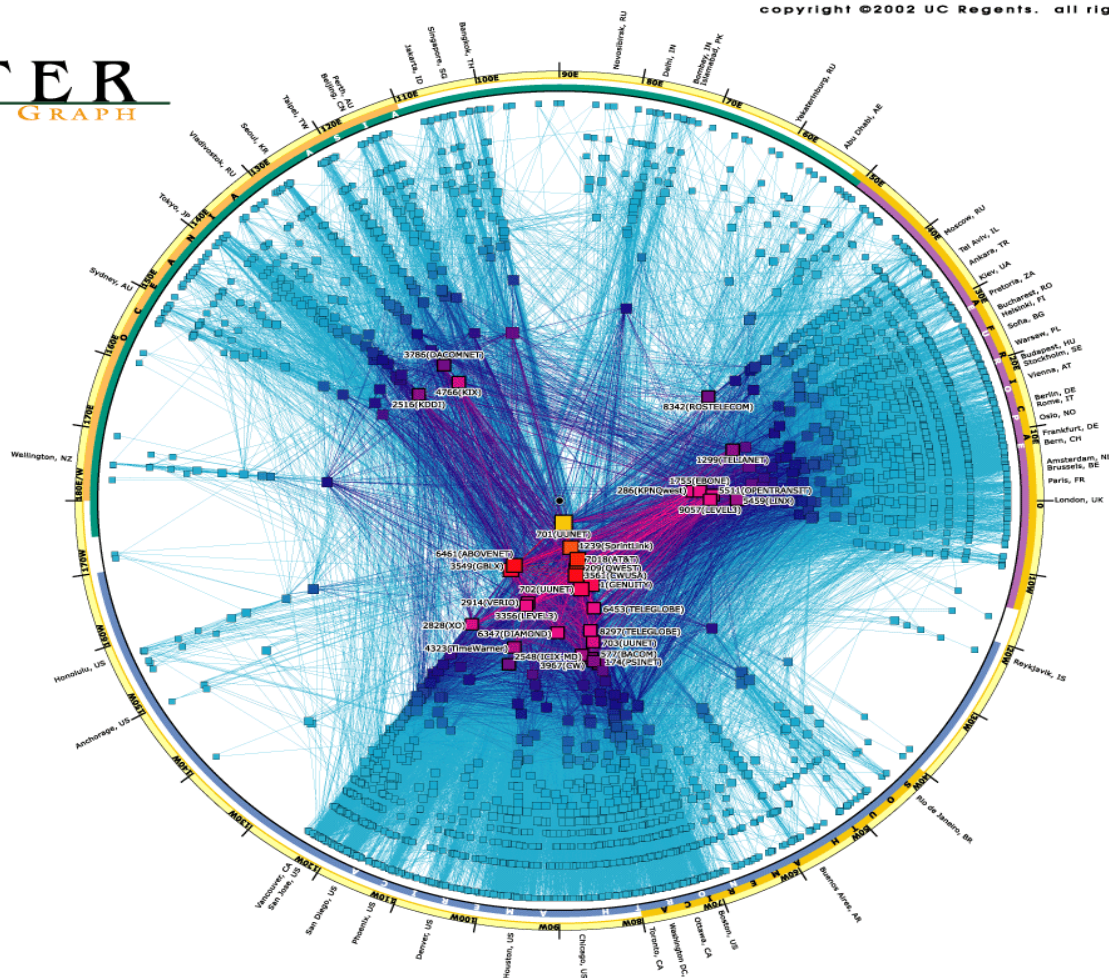
859

1030

1202

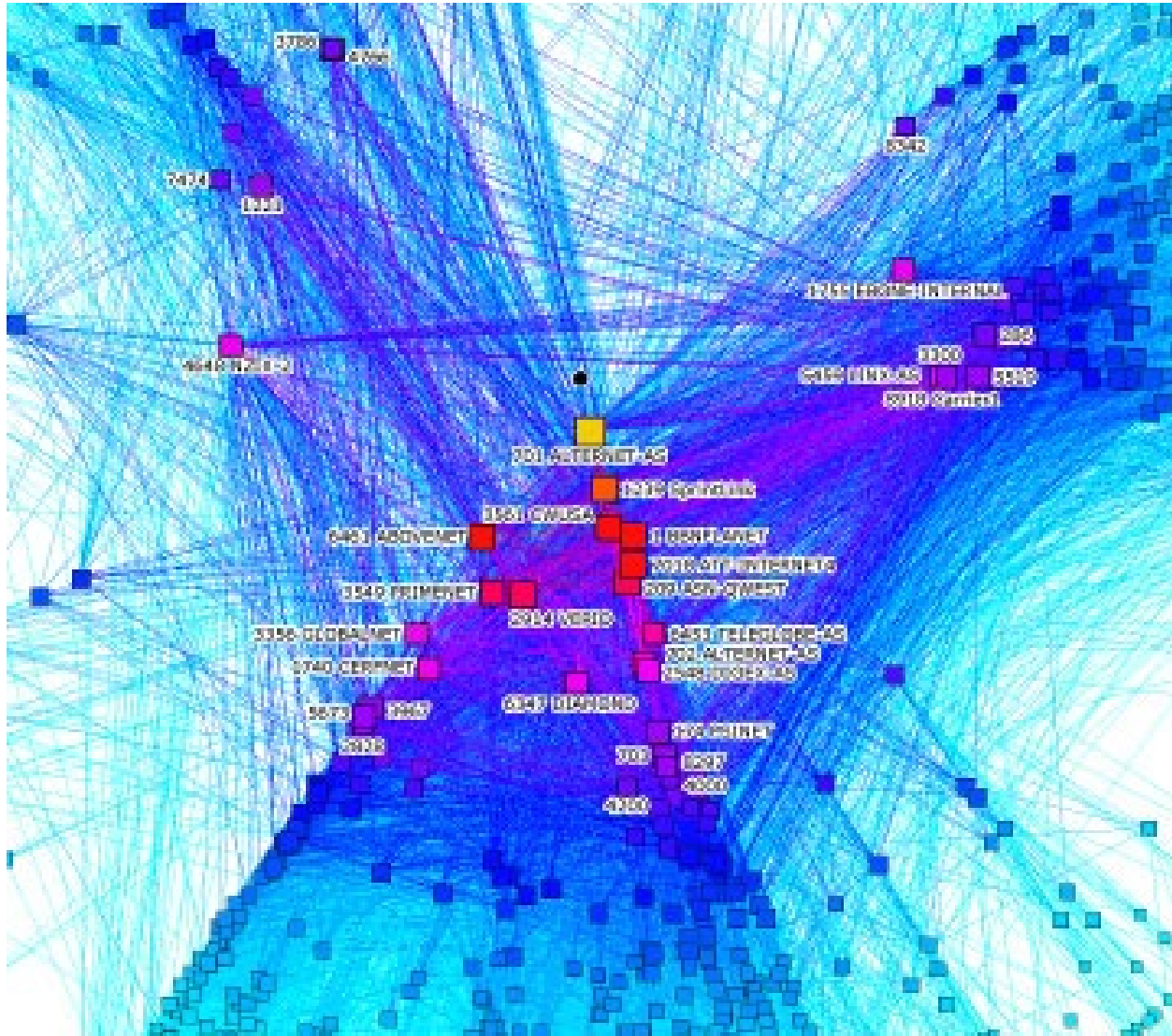
1374

1546



number of cases, 2003-2004

skitter is supported by DARPA NCI Cooperative Agreement N66001-98-2-8922, NSF ANIR Grant NCR-9711092 and CAIDA members

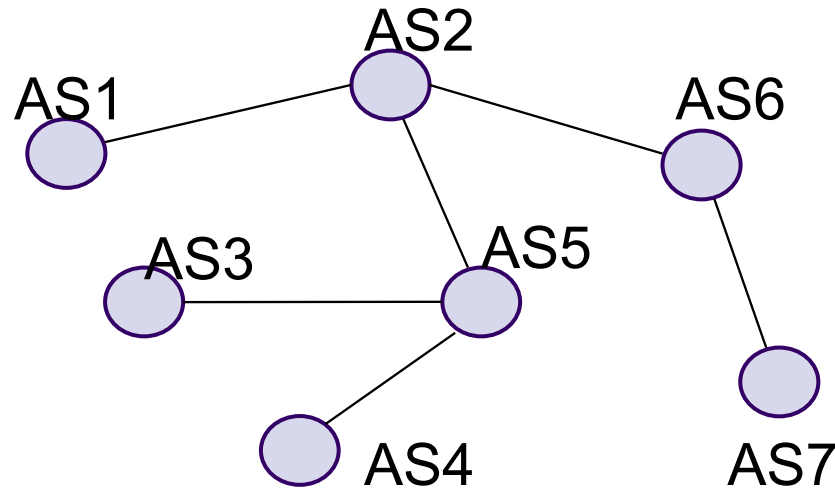




EGP Requirements

- Scalability to global Internet
 - Provide connectivity at global scale
 - Link-state does not scale
 - Should promote address aggregation
 - Fully distributed
- EGP path selection guided by policy rather than path optimality
 - Trust, peering arrangements, etc
 - EGP should allow flexibility in choice of paths

Border Gateway Protocol v4



- BGP (RFC 1771) an EGP routing protocol to exchange network reachability information among BGP routers (also called *BGP speakers*)
- Network reachability info contains sequence of ASs that packets traverse to reach a destination network
- Info exchanged between BGP speakers allows a router to construct a graph of AS connectivity
 - Routing loops can be trimmed
 - Routing policy at AS level can be applied

BGP Features



- BGP is *path vector protocol*: advertises sequence of AS numbers to the destination network
- Path vector info used to prevent routing loops
- BGP enforces policy through selection of different paths to a destination and by control of redistribution of routing information
- Uses CIDR to support aggregation & reduction of routing information